Statistical Foundations of Machine Learning

Fast Paced TA Session – Problem Sheet

6.30 – 7.30 pm, 24th Oct 2026

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1. Normal Random Variables

Problem 1.1 - Covered in previous TA session

If X is a normal random variable with mean $\mu = 3$ and variance $\sigma^2 = 16$, find:

(a) $P\{X < 11\}$

(b) $P\{X > -1\}$

(c) $P\{2 < X < 7\}$

Problem 1.2

Data from the National Oceanic and Atmospheric Administration indicate that the yearly precipitation in a certain coastal region is normally distributed with a mean of 12 inches and a standard deviation of 3 inches.

(a) Find the probability that the total precipitation during the next 2 years will exceed 25 inches.

(b) Find the probability that next year's precipitation will exceed that of the following year by more than 3 inches.

Assume that the precipitation totals for the next 2 years are independent.

2. Sample Mean and Central Limit Theorem

Problem 2.1

Approximate the probability that the sum of 16 independent uniform $(0, 1)$ random variables exceeds 10.

Problem 2.2

The amount of time that a certain type of battery functions is a random variable with mean 5 weeks and standard deviation 1.5 weeks. Upon failure, it is immediately replaced by a new battery. Approximate the probability that 13 or more batteries will be needed in a year (take 1 year = 52 weeks).

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Problem 2.3

The lifetime of a certain electrical part is a random variable with mean 100 hours and standard deviation 20 hours. If 16 such parts are tested, find the probability that the sample mean is:

(a) less than 104 hours;

(b) between 98 and 104 hours.

3. Maximum Likelihood Estimation (MLE)

Problem 3.1

Let $X_1, X_2, \ldots, X_n$ be an independent and identically distributed (i.i.d.) random sample from an exponential distribution with unknown rate parameter $\lambda > 0$, whose probability density function is given by:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Determine the maximum likelihood estimator (MLE) of λ.

Problem 3.2

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. random sample from a Laplace distribution with unknown location parameter $\mu \in \mathbb{R}$ and known scale parameter $b = 1$, whose probability density function is given by:

$$f(x;\mu) = \frac{1}{2} e^{-|x-\mu|}, \quad x \in \mathbb{R}$$

Determine the maximum likelihood estimator of the location parameter μ.

Problem 3.3

Let $X_1, \ldots, X_n$ be a sample from a normal μ, σ^2 population. Determine the maximum likelihood estimator of σ^2 when μ is known. What is the expected value of this estimator?

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